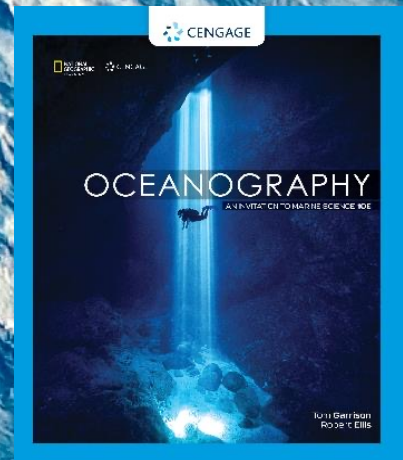


SIO 30 – The Oceans

Tiago Carrilho Biló

SIO Physical
Oceanographer
specialized in large scale
ocean circulation.

**Suggested reading: Garrison & Ellis
Oceanography: An Invitation to Marine Science**



Summer 2022 - Course format

- Introduction to Oceanography
- Study topics will be presented in lectures and discussed sessions
 - Week 1: Marine Geology
 - Week 2: Marine Chemistry
 - Week 3: Physical Oceanography
 - Week 4: Biological Oceanography
- You will be tested 5x along the course
 - Weekly homework tasks (4 in total): 60% of the grade (groups up to 5 people)
 - Final exam (closed book): 40% of the grade (07/28/2022)
- Homework and exam questions: Based on lectures and discussions (suggested reading is strongly recommended but not required)
- Suggested reading access: UCSD library or Redshelf module (ebook access)
- Extra 2 pts for those with active participation in class and discussion sessions

Summer 2022 - Course logistics

- Instructor: Tiago Carrilho Bilo (tcarrilhobilo@ucsd.edu); Office: SIO MESON Building Room 204
- TA: Michelle Guraieb (mguraieb@ucsd.edu)
- Don't hesitate in contacting us. We will do our best to solve any problems, concerns, answer question, or simply discuss oceanography. We are here to help you!
- All materials will be uploaded in Canvas



Lecture Session 1

1. Importance of ocean sciences;
2. Basic facts about our oceans and oceanography;
3. The origin of the Earth and Oceans.

Why study the oceans?

Yes, it is important!

Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;



Salt production



Tom Garrison



Why study the oceans?

A few examples:

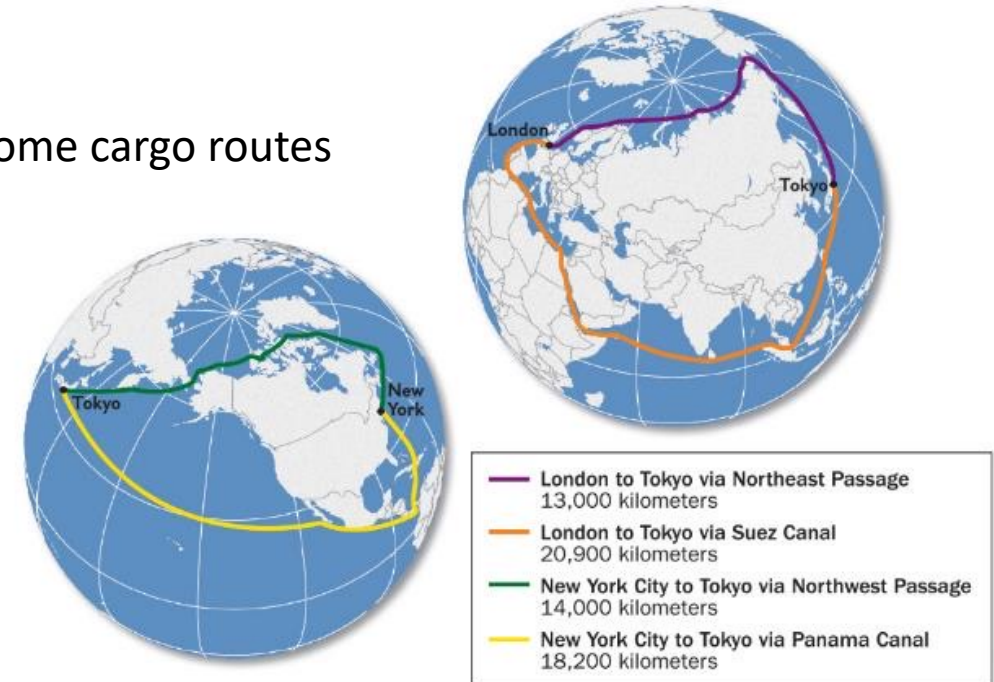
1. Resources: food, minerals, pharmaceuticals, etc;

2. Transportation;

Panama Canal lock gates



Some cargo routes



Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;
2. Transportation;
3. Recreation;



Why study the oceans?

A few examples:

1. Resources: food, minerals, pharmaceuticals, etc;
2. Transportation;
3. Recreation;
4. Biodiversity;



Kelp forest views from below the ocean surface. Credit: Andrew b Stowe

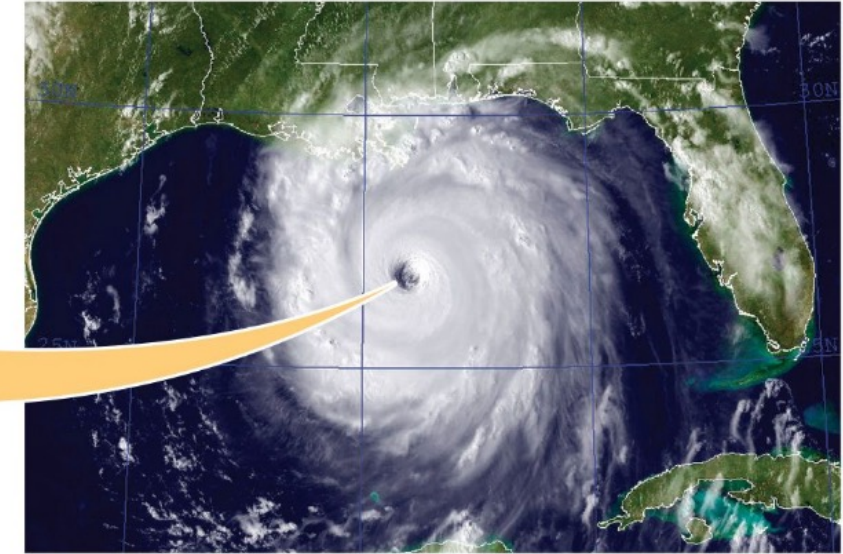
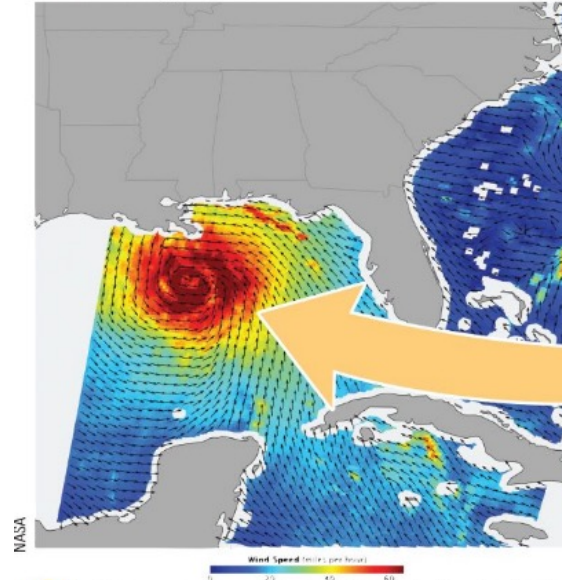
Why study the oceans?

A few examples:

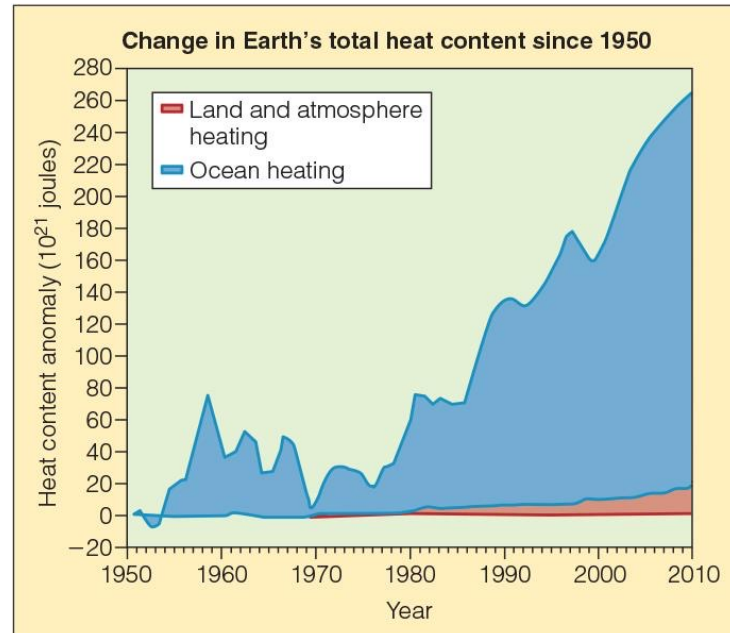
1. Resources: food, minerals, pharmaceuticals, etc;
2. Transportation;
3. Recreation;
4. Biodiversity;
5. Import from local weather and conditions to global climate

Nearly half of the planet's inhabitants live within 240 kilometers (150 miles) of a coastline (very close to the ocean)

Hurricane Katrina August 28, 2005



a Katrina as a category 5 storm hours before landfall on 28 August 2005. The storm is relatively compact with strong winds confined to about 500 kilometers (310 miles) from edge to edge. Katrina's overall shape and well-formed eye is easily seen.



Oceans slow down global warming by sequestering a lot of heat from our atmosphere. Consequence
Example: man-forced sea level rise.



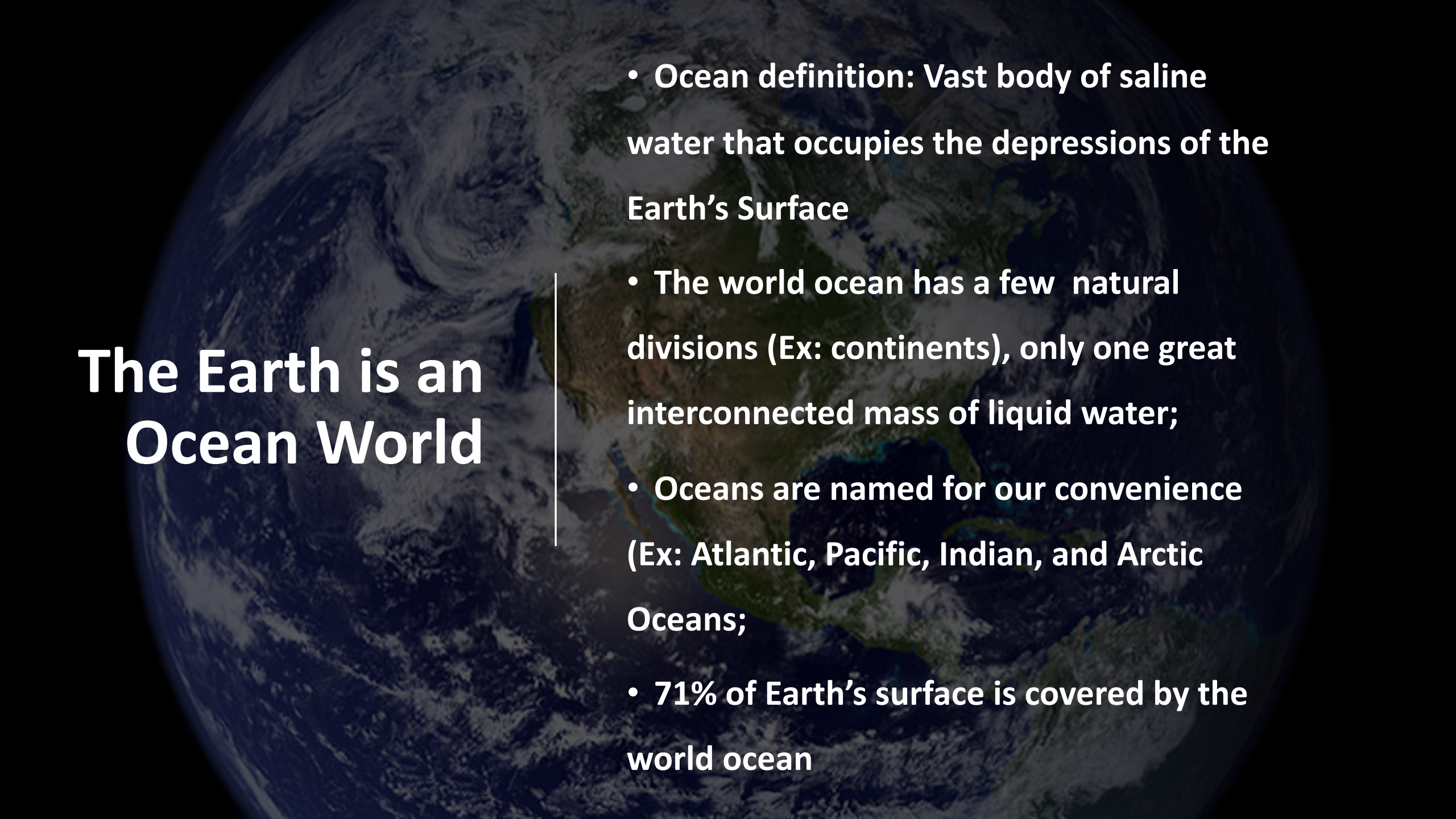
Studying the Oceans: Oceanography

Oceanography (or marine science) is the scientific study of the ocean, its associated life-forms, and its bordering lands

- Marine science draws on many disciplines: geology, physics, biology, chemistry, astronomy, atmospheric science, anthropology, ecology, computer science, applied mathematics, and engineering.
- As a marine scientist, we specialize in one area of research, but we must be familiar with related disciplines and appreciate the linkages between them.

Oceanography professionals

- Marine geologists
- Physical oceanographers
- Chemical oceanographers
- Climate specialists
- Marine biologists
- Marine engineers



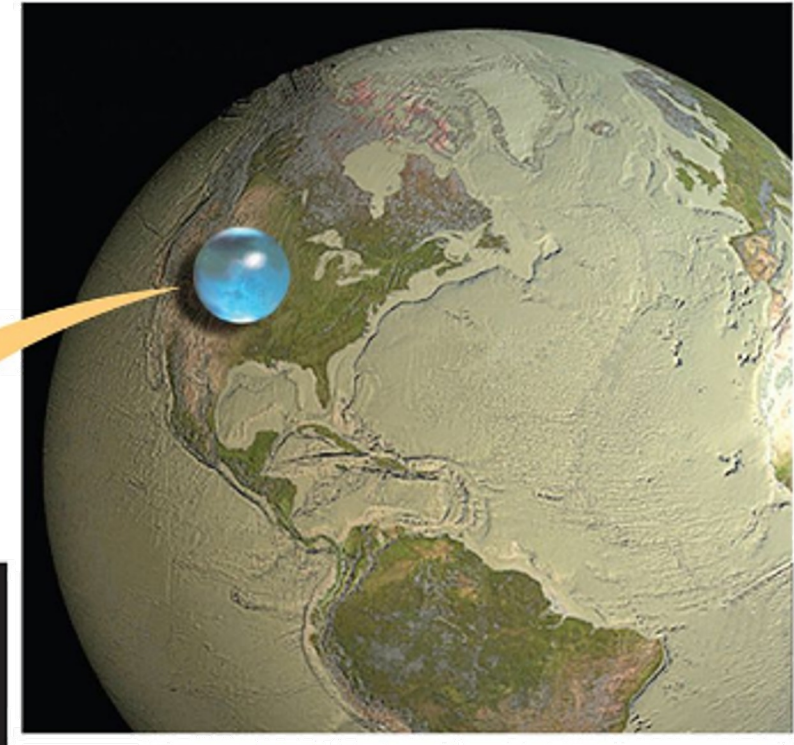
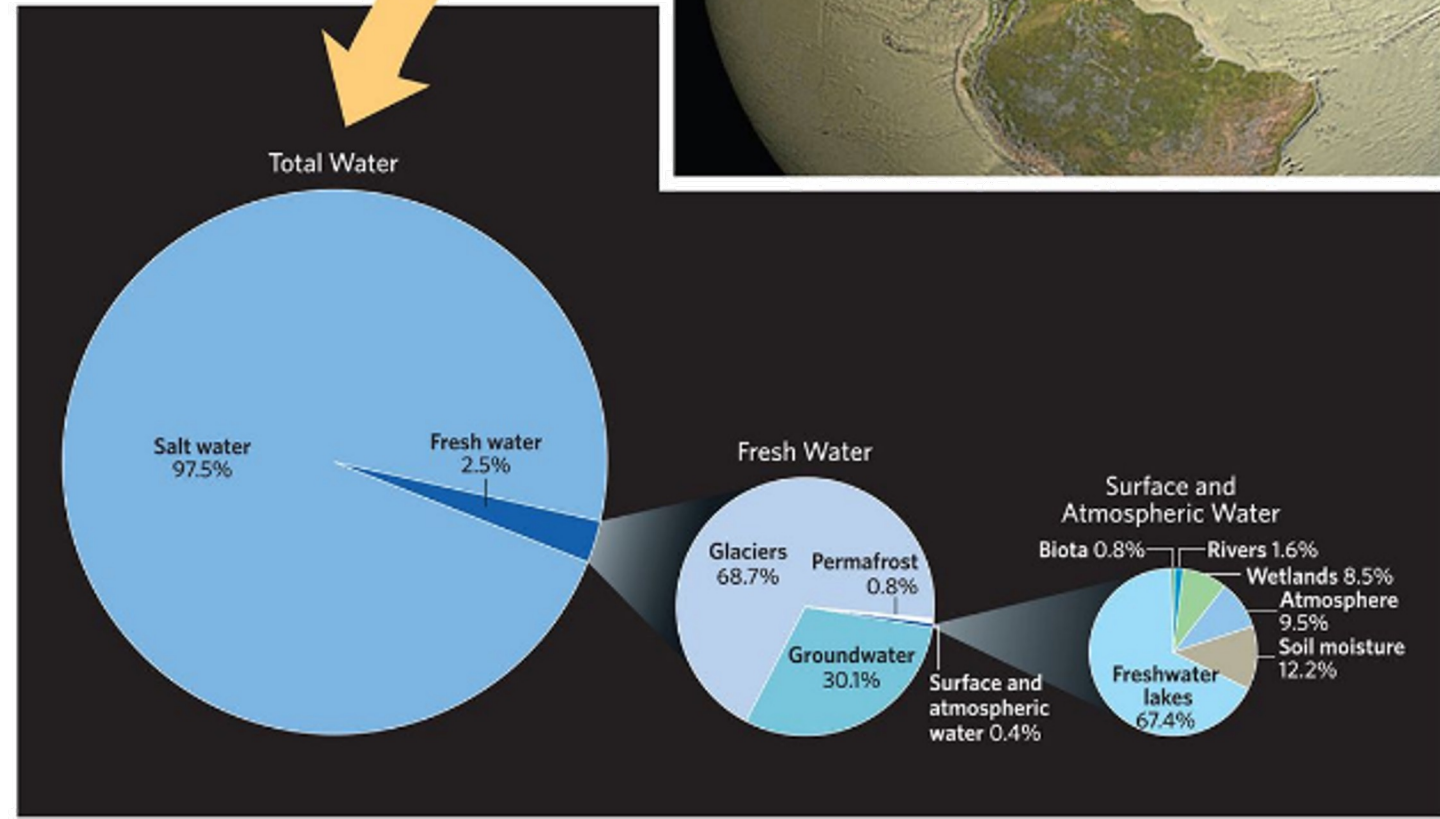
The Earth is an Ocean World

- Ocean definition: Vast body of saline water that occupies the depressions of the Earth's Surface
- The world ocean has a few natural divisions (Ex: continents), only one great interconnected mass of liquid water;
- Oceans are named for our convenience (Ex: Atlantic, Pacific, Indian, and Arctic Oceans;
- 71% of Earth's surface is covered by the world ocean

Getting some perspective:

Relative amount of water on the Earth's surface

More than 97% of the water lies in the ocean. If all water at Earth's surface were gathered up into a sphere, its diameter would measure only 1,380 km (860 miles)



Getting some perspective:

Relative amount of water on the Earth's surface

Those are huge numbers!

On a human scale, the ocean is mighty impressive



Statistics for the World Ocean

Total area:	331,441,932 square kilometers	127,970,445 square miles
Total volume:	1,303,155,354 cubic kilometers	312,643,596 cubic miles
Total mass:	1.41 billion billion metric tons	1.55 billion billion tons
Average depth:	3,682 meters	12,081 feet
Greatest depth:	10,994 meters	36,070 feet

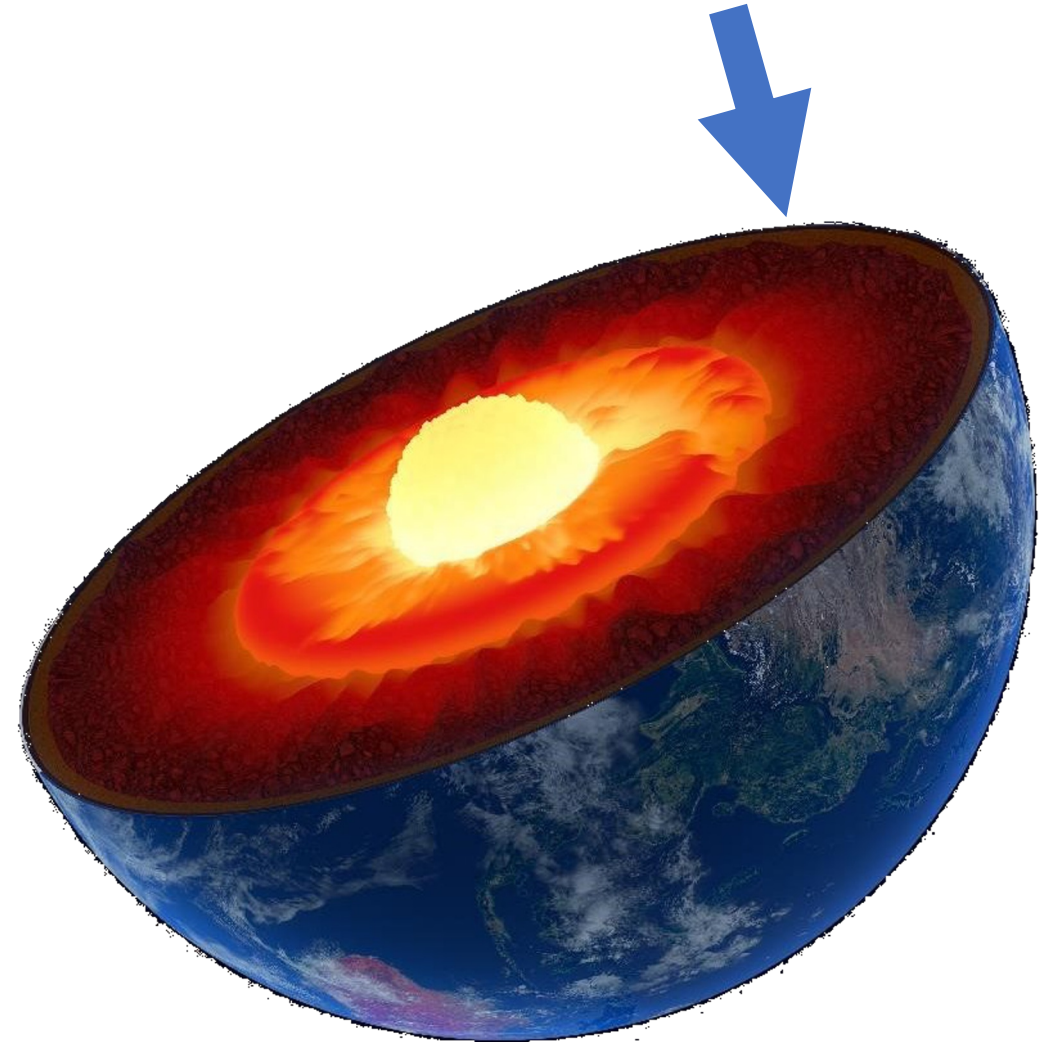
Getting some perspective:

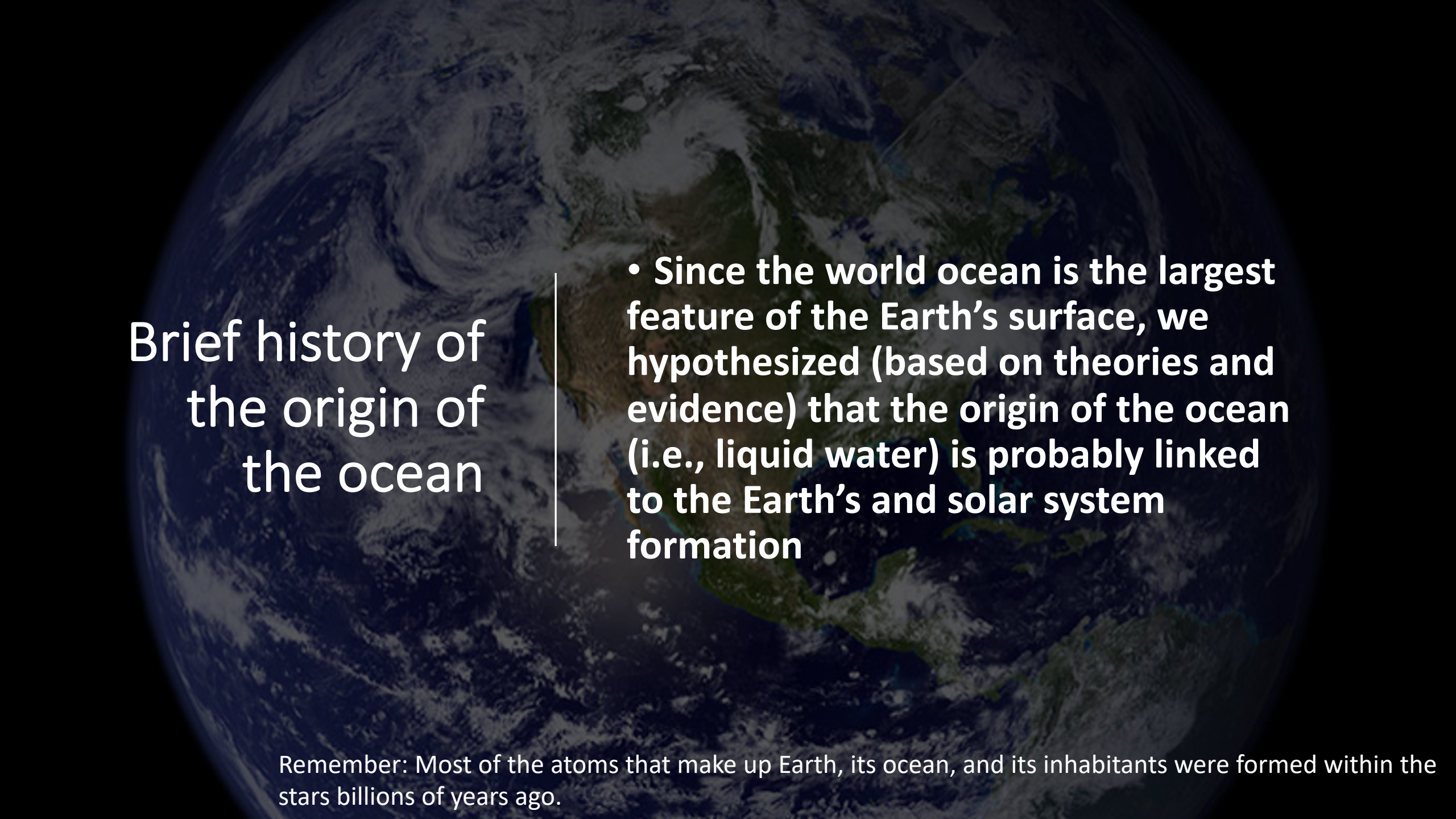
Relative amount of water on the Earth's surface

On a planetary scale, not so much

- The average depth is a tiny fraction of Earth's radius.
- Accounts for only slightly more than 0.02% of Earth's mass (0.13% of its volume).
- Much more water is trapped within the Earth's hot interior than exists in its ocean and atmosphere.

Ocean is over a
very thin crust





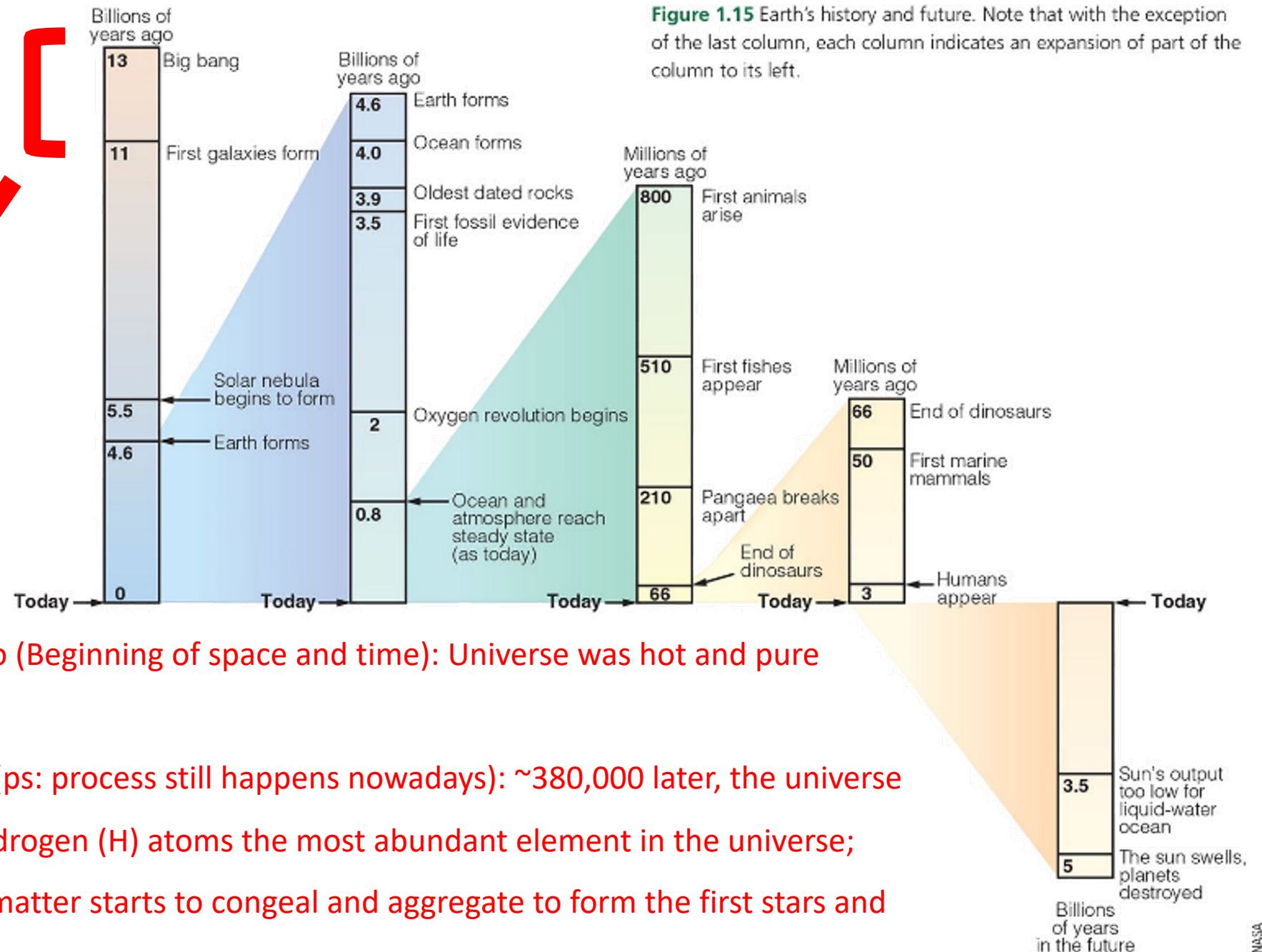
Brief history of the origin of the ocean

- Since the world ocean is the largest feature of the Earth's surface, we hypothesized (based on theories and evidence) that the origin of the ocean (i.e., liquid water) is probably linked to the Earth's and solar system formation

Remember: Most of the atoms that make up Earth, its ocean, and its inhabitants were formed within the stars billions of years ago.

Universe and Solar System Formation

The beginning of everything

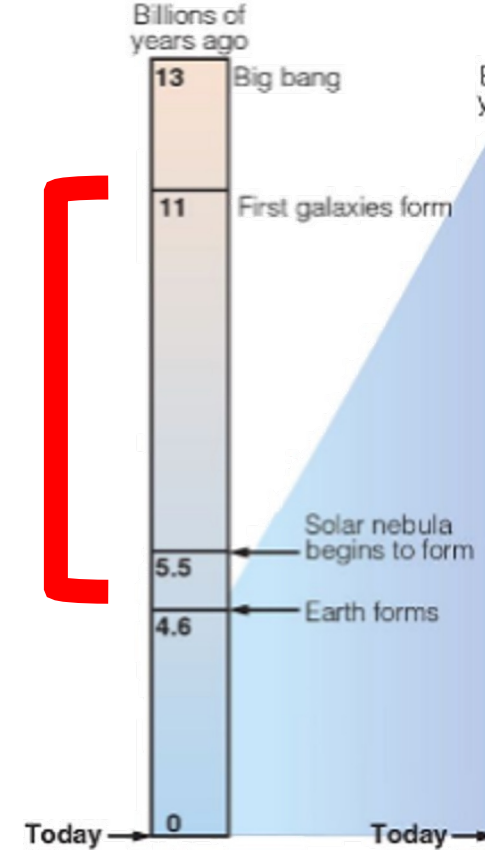


1. Big Bang ~13 Billions of years ago (Beginning of space and time): Universe was hot and pure “raw” energy;
2. Universe expansion and cooling (ps: process still happens nowadays): ~380,000 later, the universe became cool enough to form Hydrogen (H) atoms the most abundant element in the universe;
3. In the course of ~1 billion years matter starts to congeal and aggregate to form the first stars and galaxies

Universe and Solar System Formation

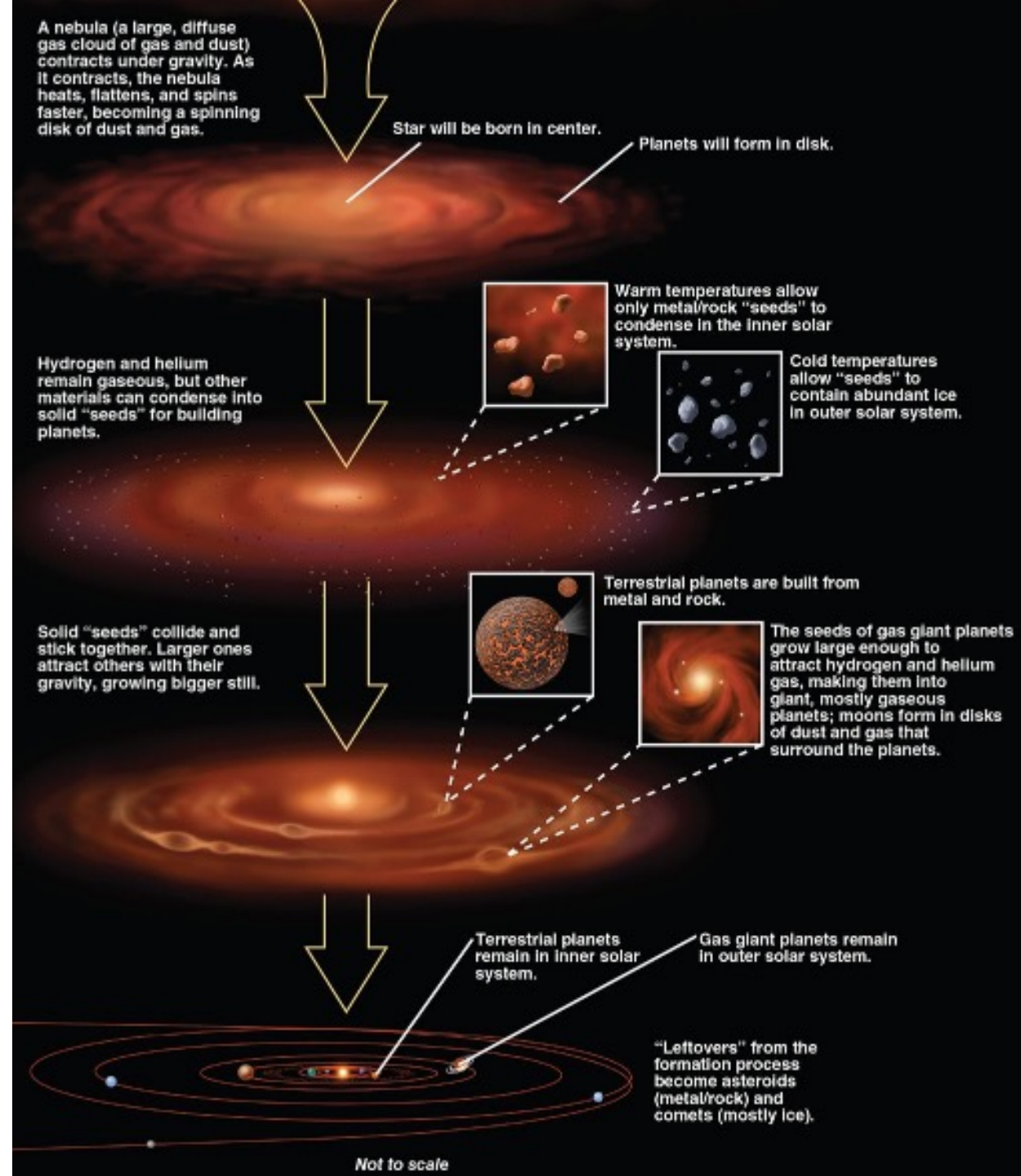
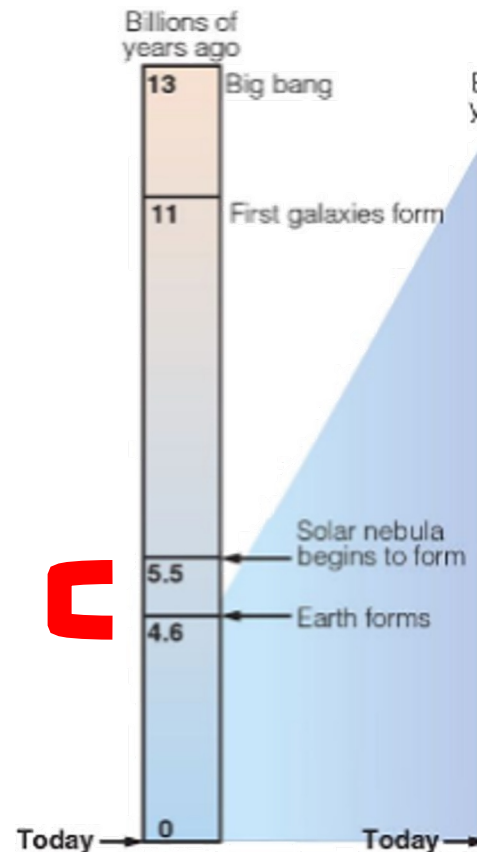
From heavier elements to Earth: Stars produced all matter we find today (and continue to do so)

- Nuclear Fusion inside stars: Mainly produces Helium (He), however stars can produce elements as heavy as Carbon (C);
- Supernovas and the death of stars: BY the final stages of the life cycle of stars they explode forming supernovas. During such explosive events, the large amount of energy liberated are partially used to create even heavier elements like Iron (Fe) and Nickel (Ni), forming the building blocks of more complex molecules (water in the universe probably originated during this period).



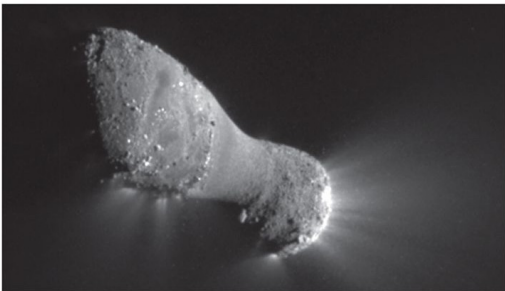
Our Sun and Earth formed from a dying star via the process of **accretion** ➔

The accretion process that formed our Solar System lasted ~30-50 millions of years



4.6 Billions of year ago Earth finally formed (our ocean has a home)

1. At first, Earth was very hot: Only gases (including water vapor) and molten rock;
2. Cooling and the origin of liquid water: Rocks at the surface started to form a solid crust;
3. Water vapor trapped under the crust was released to the atmosphere via volcanic activity (outgassing). Constant water condensation helped cooling the Earth's surface and liquid accumulated over the crust. **This accumulation originated large bodies of water such as the ocean.**
4. **Liquid water at the surface (and life) only exist on Earth because our especial position in the Solar System (aka Habitable Zone).**
- 5 Asteroids and comets probably were external water sources to Earth



Asteroid

NASA/JPL-Caltech/UMD



Tom Gernsheim

Outgassing

Ocean forms around 4.0 Billions of years ago

